AGROFILOS

Automatic Multipurpose Pesticide Sprayer

Unmanned Ground Vehicle (UGV) for Agricultural Use

Developer	Ayushman Sahoo
Institution	DAV Public School, Kalinga Nagar, Bhubaneswar, Odisha
Stage	Prototype — Self Tested
Category	Agricultural Technology / Robotics
Document Date	April 2026

🏆 Awards & Recognition

CBSE National Science Exhibition 2023 • IdeaNest 2.0 Startup Bootcamp • Dr. APJ Abdul Kalam Ignited Minds Award 2022 • Sahodaya Science Exhibition

1. Executive Summary

AGROFILOS is India's first automatic multipurpose pesticide sprayer designed as an Unmanned Ground Vehicle (UGV) to reduce the physical burden and health risks faced by farmers during crop spraying operations. Developed by Ayushman Sahoo as an innovation project, AGROFILOS addresses the critical gap between expensive agricultural drones and hazardous manual sprayers by providing an affordable, easy-to-use, remotely operated UGV for Indian farming conditions.

The system enables farmers to spray pesticides, herbicides, insecticides, and fertilizers without physical strain, exposure to harmful chemicals, or risk from venomous insects. It is operable via an RF remote from over 100 metres, with independently adjustable speed and pump pressure.

Key Innovation

AGROFILOS bridges the affordability gap in Indian agriculture technology. At a fraction of the cost of agricultural drones (₹4–5 lakh), it delivers superior functionality including unlimited run time, easy operation, large pesticide capacity, and full adjustability for crops of any height — making it accessible to the 82% of Indian farmers who are small and marginal.

Core Specifications at a Glance

Parameter	Compact Mode	Full Extension
Height	5 feet (152 cm)	9 feet (274 cm)
Length	100 cm	6.3 feet (192 cm)
Width (Base)	100 cm	5 feet (152 cm)
Sprinkler Pipe	~5 feet	8.5 feet (259 cm)
Coverage Area	—	110 sq. ft. per pass
Tank Capacity	16 litres	16 litres
Remote Range	100+ metres	100+ metres
Battery Runtime	1+ hour	1+ hour
Pump Pressure	130 PSI	130 PSI
Pump Flow Rate	6 L/min	6 L/min

2. Problem Statement & Background

2.1 The Scale of the Problem

Indian agriculture is heavily dependent on human labour for pesticide and fertilizer application. With 82% of India's farming households being small and marginal farmers, access to advanced agricultural technology remains severely limited. The existing landscape presents two unsatisfactory extremes:

Aspect	Agricultural Drones	Manual Sprayers
Cost	₹4–5 lakh (unaffordable for 82% of farmers)	₹1,000–10,000 but uses petrol; recurring costs
Operation	Requires technical training; complex to operate	Farmer must walk the entire field manually
Health Risk	Low for farmer, but misuse risks exist	Direct chemical exposure; asthma & skin diseases
Physical Strain	None for farmer	Heavy backloads; fatigue; back injuries
Snake/Pest Risk	None — airborne	High — farmer walks through crop rows
Runtime	~20 min flight time per charge	Unlimited but exhausting
Capacity	Low (small pesticide tanks)	Moderate but heavy when full
Environment	Clean (electric)	Petrol fumes; not eco-friendly

2.2 Existing Challenges

The pesticide application challenge in India is compounded by several structural factors:

- 70% of rural Indian households depend primarily on agriculture for their livelihood.
- 82% of India's farmers are small and marginal, with limited capital for machinery.
- Back-loaded sprayers cause musculoskeletal disorders from repeated use on rough terrain.
- Petrol-powered sprayers increase input costs and carbon emissions.
- Walking through crop rows significantly increases risk of snake and venomous insect bites.
- Prolonged pesticide exposure is linked to respiratory illness, skin disease, and chronic poisoning.

3. Project Inspiration & Objectives

3.1 Inspiration

AGROFILOS was conceived as a direct response to the daily hardships observed in Indian farming communities. The project stems from understanding that while agricultural technology has advanced globally, cost and complexity barriers prevent its adoption at the grassroots level in India.

The inspiration was rooted in a fundamental observation: India's farmers — who feed a billion people — are still walking through fields with chemical tanks strapped to their backs, exposed to health hazards and physical exhaustion that could be eliminated with appropriate technology at the right price point.

Driving Question

"Can we build a machine that removes the farmer from the hazardous chemical environment entirely, is affordable enough for a small-scale farmer, and is simple enough to operate without any technical expertise?"

3.2 Objectives

- Develop a machine that eliminates manual labour and physical strain during pesticide application.
- Enable farmers to control spraying operations remotely from a safe distance (100+ metres).
- Design an adjustable and extendable structure that works for crops of any height — from ground-cover to sugarcane.
- Achieve affordability that makes the machine accessible to small and marginal farmers.
- Build an eco-friendly, electric-powered alternative to petrol sprayers.
- Maximise pesticide coverage efficiency and uniformity through a multi-sprinkler system.

4. Solution Overview

4.1 Three-Component Architecture

AGROFILOS is divided into three independently functional but integrated sub-systems:

🚜 Main Body	🖥️ Main Controlling Unit (MCU)	💧 Sprinkling System
Telescopic PVC frame with triangular base structure, 4 wheels (2 motorised), spring suspension, and 16L pesticide container. Extendable in all dimensions.	Arduino Nano-based control board with RF receiver, PWM motor driver, pump controller, battery management, voltage sensor, and status LEDs.	Independent telescopic sprinkler pipe with 7 nozzles whose flow and pressure can be controlled individually. Adjustable height regardless of main body.

4.2 Novel Features

- Fully telescopic frame — adjustable height, width, and length to suit any crop type.
- Independent sprinkler pipe — height adjustable separately from the main body for precise nozzle positioning.
- 7-nozzle sprinkler array — each nozzle controllable independently; uniform coverage of 110 sq. ft. per pass.
- Triangular base structure — provides structural rigidity and stability on uneven agricultural terrain.
- Spring suspension on all 4 wheels — absorbs field irregularities without tilting the pesticide tank.
- Wireless RF remote (315/433 MHz) — speed and pump pressure control from 100+ metres distance.
- Water level indicator system — 4-LED array on MCU blinks at increasing rates as fluid level drops.
- Battery charger with auto cut-off — charges via standard 220V outlet; prevents overcharging.

5. System Architecture

5.1 Mechanical Design

The mechanical architecture of AGROFILOS is built around a telescopic PVC pipe system with flexible coupling joints at every extension point. The triangular cross-section of the base frame distributes load symmetrically and widens the footprint as the machine is raised, maintaining stability at all heights.

Sub-System	Compact Configuration	Extended Configuration
Frame Height	5 feet (152 cm)	9 feet (274 cm)
Base Width	100 cm	5 feet (152 cm)
Frame Length	100 cm	6.3 feet (192 cm)
Sprinkler Pipe	~5 feet	8.5 feet
Storage Footprint	Small — fits in tight spaces	Expanded for operation

5.2 Electrical & Control Architecture

The control system uses a dual-Arduino architecture — one Arduino Nano on the handheld transmitter (remote) and one on the Main Controlling Unit (MCU) on the machine. RF modules (RadioHead library, 315/433 MHz) handle wireless communication, with a range exceeding 100 metres.

The communication protocol encodes the combined state of speed (4 levels: 0, 153, 205, 255 PWM) and flow (4 levels: 0, 75, 125, 255 PWM) into single character codes (a–H), transmitted as single-byte RF packets. Directional commands (i–p) and a stop command (T) handle movement. This compact encoding reduces transmission overhead and improves reliability.

RF Code	Speed (PWM)	Flow/Pump (PWM)	Action
a	0	0 (off)	Stationary, pump off
b	153 (~60%)	0 (off)	Slow, pump off
c	205 (~80%)	0 (off)	Medium, pump off
d	255 (100%)	0 (off)	Full speed, pump off
e	0	75 (low)	Stationary, low spray
f–h	153–255	75 (low)	Moving with low spray
A–D	0–255	125 (mid)	Various speed, mid spray
E–H	0–255	255 (full)	Various speed, full spray
i / j	—	—	Forward / Backward
k / l	—	—	Right / Left turn (tank steer)
m–p	—	—	Diagonal movements (WD/WA/SD/SA)
T	—	—	Stop all motors

5.3 Signal Flow

The complete signal flow from operator to field operation is:

- Operator presses button on RF remote → Arduino Nano (Transmitter) reads digital input.
- Transmitter encodes speed + flow state into single character → RF module broadcasts at 315/433 MHz.
- MCU Arduino Nano (Receiver) receives packet via RadioHead library → decodes character.
- MCU sets PWM values on ENA/ENB pins of motor driver → controls drive motor speed.
- MCU sets PWM value on pump control pin → regulates pump pressure and flow.
- Motor driver (Cytron MDD3A / L298N) drives left and right wheel motors independently for steering.

6. Hardware Components

6.1 Main Controlling Unit (MCU) — Electronics

Component	Function	Specification	Notes
Arduino Nano	Main controller for MCU; decodes RF signals, drives motors and pump	ATmega328P, 16 MHz	2× (one per unit)
RF 315/433 MHz Module (RadioHead)	Wireless communication between remote and machine	100m+ range	RH_ASK library
Dual DC Motor Driver	Drives left and right wheel motors independently	3A per channel	Cytron MDD3A
12V High-Torque DC Motors	Rear wheel drive; provides traction on rough terrain	125 RPM, 150 N·cm	Orange brand
High-Pressure Water Pump	Pesticide delivery with variable pressure	130 PSI, 6 L/min	12V DC
P90NF03L N-MOSFET	PWM-based pump speed/pressure regulator	30V, 90A	Gate-controlled
12V Lead-Acid / Li-Ion Battery	Powers entire system	12V nominal	1+ hr runtime
Battery Charger Module	Charges via 220V AC; auto cut-off when full	TP4056 / dedicated	Onboard
Voltage Divider (R1=30kΩ, R2=7.5kΩ)	Battery voltage sensing for low-battery alert	Scales 12V to 5V	Feeds Arduino A0
Water Level Sensor	Monitors pesticide level in tank	Analog resistive	Feeds Arduino A5
Status LEDs (×6)	Power, battery, pairing, water level indication	Various colours	Onboard MCU

6.2 RF Transmitter (Remote Control) — Electronics

Component	Function	Notes
Arduino Nano	Reads button inputs; encodes and transmits RF packets	ATmega328P
RF 315/433 MHz TX Module	Broadcasts control signals to MCU receiver	RadioHead RH_ASK
8× Push Buttons	Directional (W/A/S/D), speed+, speed−, flow+, flow−	Active LOW
1× Toggle Switch	Clutch — enables speed/flow adjustment mode	Digital input
Voltage Divider (R1=26kΩ, R2=6.6kΩ)	Remote battery voltage sensing	Min 6.6V threshold
8× LED Indicators (S0–S3, F0–F3)	Visually shows current speed and flow level selected	Active HIGH

1× Indicator LED	Blinks when remote battery is low	indi pin 11
9V Battery	Powers remote unit	Standard PP3

6.3 Structural Materials — Bill of Materials

Sl.	Description	Qty.	Rate (₹)	Amt. (₹)	Running Total (₹)	Notes
1	3/4" cPVC male adapter	16	26	416	416	Frame joints
2	3/4" × 1/2" cPVC female reducer	4	10	40	456	
3	3/4" cPVC pipe (15 ft)	1	27.5	412	868	Sprinkler pipe
4	1/2" cPVC pipe (5 ft)	1	18.5	93	960	
5	1.5" (40mm) PVC pipe (13'6")	1	18	243	1203	Main frame
6	2" (50mm) PVC pipe (2 ft)	1	24	48	1251	
7	1.5" (40mm) PVC Tee	4	22	88	1339	
8	3/4" cPVC Tee	4	24	96	1435	
9	2" × 1.5" reducer	2	20	40	1475	
10	1/2" cPVC end cap	2	8	16	1491	
11	cPVC solvent solution	1	135	130	1621	Pipe adhesive
12	Spring (small)	4	50	200	1821	Suspension
13	Sprinkler nozzle (×7 set)	1	150	150	1971	
14	Flexible tube (5 ft)	1	10	50	2021	Pump outlet
15	Gray spray paint	1	300	300	2321	
16	Black spray paint	2	247.5	495	2816	
17	Teflon tape (×4 rolls)	4	20	80	2896	
18	GI 3/4" 3-way elbow	4	60	240	3136	
19	4" (110mm) PVC pipe (3'6")	1	50	175	3311	Wheel axle
20	Additional PVC/uPVC pipes & tees	—	—	679	3990	Various diameters

–24						
25–32	GI rod, bolts, nuts, washers, springs, epoxy	—	—	773	4763	Hardware

Total Structural Materials Cost: ₹4,763

7. Software Architecture

7.1 Firmware Overview

AGROFILOS uses two Arduino Nano microcontrollers running custom firmware written in Arduino C/C++ using the RadioHead (RH_ASK) library for RF communication. The firmware reached version 4.0 (Transmitter_4.0.ino and Mains_4.0.ino), superseding earlier versions based on VirtualWire and Bluetooth (HC-05).

File	Role	Key Responsibilities
Transmitter_4.0.ino	RF Remote Firmware	Reads button inputs, encodes speed+flow state as single char, broadcasts RF packet; monitors remote battery; drives LED indicators
Mains_4.0.ino	MCU (Machine) Firmware	Receives RF packets, decodes commands, sets motor PWM, controls pump PWM, monitors machine battery and water level, manages status LEDs and pairing

7.2 RF Communication Protocol

The transmitter encodes the combined state of motor speed and pump flow into a single ASCII character before transmission. This compact encoding scheme minimises packet size and transmission time:

	Flow: 0 (off)	Flow: 75 (low)	Flow: 125 (mid)	Flow: 255 (high)	Notes
Speed: 0	a	e	A	E	Stationary
Speed: 153	b	f	B	F	Slow (~60%)
Speed: 205	c	g	C	G	Medium (~80%)
Speed: 255	d	h	D	H	Full speed

7.3 Key Firmware Features — MCU (Mains_4.0)

- Pairing mode: MCU blinks pair LED until paired remote sends 'T' (all buttons released); pair LED turns off after first successful pairing.
- Tank level monitoring: Water level sensor on A5 read continuously; 4 LEDs (led1–led4) blink at different rates based on fluid level — rapid blink = nearly empty.
- Battery voltage sensing: Voltage divider (R1=30kΩ, R2=7.5kΩ) on A0 measures battery level; indi LED activates below 10.2V threshold.
- Tank steer differential drive: Turn radius controlled by applying reduced PWM (spid−100) to the inner wheel while outer wheel runs at full spid.
- Motor direction control: Full H-bridge control via rForward/rBackward and lForward/lBackward pins with independent left/right channels.

7.4 Development History

The firmware evolved through multiple versions as the RF communication technology was refined:

Version	RF Technology	Transmitter File	Main Unit File
v1.0	VirtualWire (433 MHz)	Transmiter2.0.ino	TX 1.0 / RX 1.0 (Dummy)
v2.0	VirtualWire (refined)	Transmiter2.0.ino	Main_2.0.ino
v3.0	HC-05 Bluetooth	Transmitter_3.0.ino	Main_3.0.ino
v4.0 (Current)	RadioHead RH_ASK (315 MHz)	Transmitter_4.0.ino	Mains_4.0.ino

8. Testing & Results

8.1 Self-Testing Methodology

AGROFILOS was self-tested through a series of structured functional tests covering the control system, drive mechanism, pump and sprinkler performance, and safety features. All tests were conducted by the developer in a controlled outdoor environment.

Test Area	Test Conducted	Result
RF Communication Range	Remote operated from increasing distances; maximum reliable range measured	Functional at 100+ metres
Motor Speed Control	All 4 speed levels (0, 153, 205, 255 PWM) verified via remote	Correct PWM output confirmed on oscilloscope
Pump Pressure Control	All 4 flow levels (off, 75, 125, 255) tested with water	Even spray pattern observed at all levels
Directional Control	All 8 directions tested on flat and rough surface (W/A/S/D + diagonals)	Responsive tank-steer; all directions functional
Battery Monitoring	Battery drained below threshold; low-battery LED activation tested	indi LED activated correctly below 10.2V
Water Level Indicator	Tank filled, operated until 3 different levels; LED blink rates observed	LED blink frequency increased correctly with lower level
Pairing Mechanism	Power on sequence; pair LED blink confirmed; pairing via T command tested	Pair LED stopped blinking after first T command
Telescopic Extension	All telescopic joints extended and locked; structural integrity assessed with full tank load	Stable at all heights with 16L fluid load
Sprinkler Coverage	Spray pattern measured at full extension (8.5 ft pipe, 7 nozzles)	~110 sq. ft. even coverage confirmed
Terrain Performance	Machine driven over gravel, soil ridges, and uneven ground	Spring suspension maintained stability; no tipping

Test Summary

All primary functional tests passed successfully. The machine demonstrated stable operation across different terrain types, reliable wireless control at design range, accurate battery and water level monitoring, and uniform spray coverage across the full 110 sq. ft. spray width. No critical failures were encountered during testing.

9. Business Model

9.1 Target Market

India has approximately 146 million farm holdings, of which 82% are small and marginal (under 2 hectares). These farmers collectively represent an underserved market of over 100 million potential customers who currently rely on manual sprayers and cannot afford agricultural drones.

Canvas Area	Element	Detail
Customer Segments	**Primary**	Small and marginal farmers (under 2 ha); cooperative farming groups
Customer Segments	**Secondary**	Large farm operators; government agricultural departments; NGOs
Revenue Streams	**Core Product**	Direct sales of the AGROFILOS unit
Revenue Streams	**Ancillary**	Compatible pesticide and fertilizer supply; service contracts; replacement parts
Value Proposition	**Core**	Electric, remote-controlled, affordable, adjustable sprayer — no physical strain, no chemical exposure, no snake risk
Unique Advantage	**Vs Drone**	Lower cost, unlimited run time, no technical training needed, greater pesticide capacity
Unique Advantage	**Vs Manual**	Zero physical strain, eco-friendly, no health hazard, remote operation from 100m
Cost Structure	**Primary**	R&D, manufacturing, quality testing
Cost Structure	**Secondary**	Marketing, last-mile delivery to rural areas
Channels	**Digital**	Social media campaigns targeting farming communities
Channels	**Physical**	Local agri-dealers, village-level demonstrations, word-of-mouth

10. Challenges & Solutions

Challenge Area	Problem	Solution Implemented
RF Communication	Initial VirtualWire library was unreliable at range and prone to interference in open fields	Migrated to RadioHead RH_ASK library with 315 MHz modules; significantly improved range and reliability
Wireless Protocol Version	Bluetooth (HC-05) in v3.0 had limited range (~10m) and required pairing overhead	Switched to 315/433 MHz RF in v4.0; range increased to 100+ metres without pairing complexity
Structural Stability	Telescopic joints initially had play and wobble when fully extended with fluid load	Flexible couplings added at each telescopic junction; triangular geometry ensures self-stiffening as height increases
Terrain Navigation	Early design without suspension caused the machine to tilt on uneven field terrain	Spring suspension added to all 4 wheel mounts; absorbs ground irregularities without disturbing tank level
Pump Speed Control	Direct switching of pump motor caused uneven spray pressure	P90NF03L N-MOSFET used as PWM-controlled regulator; smooth analog control of pump flow rate
Water Level Detection	Liquid level in opaque PVC tank could not be visually assessed during operation	Analog water level sensor added at A5 pin; LED indicator array with progressive blink rate communicates level to operator
Battery Monitoring	Operator unaware of battery depletion until sudden shutdown	Voltage divider circuit added; indi LED activates when voltage drops below 10.2V minimum threshold
Wiring Protection	Exposed motor wiring susceptible to moisture and pesticide contact	Motor cables routed through hollow PVC pipe structure; MCU enclosure sealed against moisture and chemicals
Spray Coverage Uniformity	Single or dual-nozzle configurations produced uneven coverage	7-nozzle sprinkler array along 8.5 ft telescopic pipe; each nozzle independently controllable for coverage optimisation
Storage Space	Fixed-width sprayers require large dedicated storage area	Fully telescopic design; machine can be collapsed to compact form for storage in standard agricultural shed

11. Future Improvements

11.1 Hardware Upgrades

- Replace Arduino Nano with ESP32 for built-in WiFi and Bluetooth; enables OTA firmware updates and IoT monitoring of machine state.
- Upgrade to brushless motors with encoders for precise speed feedback and better torque at low speeds.
- Implement 4-wheel differential drive (all 4 wheels motorised) for better traction on muddy fields.
- Add ultrasonic/LIDAR obstacle avoidance sensors for semi-autonomous operation.
- Upgrade battery to LiFePO4 (lithium iron phosphate) for better cycle life and safety in outdoor conditions.

11.2 Control System

- Replace single-char RF protocol with LoRa for extended range (1–2 km) and bidirectional telemetry.
- Add GPS module for track recording and field mapping integration.
- Develop a smartphone app (Android) for touch-based control and real-time status monitoring via Bluetooth or WiFi.
- Implement auto-steering using field boundary GPS coordinates for straight-row navigation.

11.3 Mechanical Design

- Upgrade to aluminium alloy telescopic arms to reduce weight while maintaining structural rigidity.
- Design quick-release sprinkler nozzle system for easier maintenance and nozzle replacement.
- Add a seed-drill or soil tiller attachment interface to make the platform truly multipurpose.
- Develop injection-moulded MCU enclosure rated IP65 for dust and water resistance.

11.4 Production Targets

- Target retail price: ₹15,000–25,000 for mass-production version (compared to ₹4–5 lakh agricultural drones).
- Explore government PMKSY (Pradhan Mantri Krishi Sinchayee Yojana) subsidy eligibility for affordable farmer access.
- Establish local assembly to qualify for Made-in-India agricultural equipment incentives.

12. Events & Recognition

AGROFILOS has been presented at multiple national-level competitions and innovation events. The following records the project's competitive history:

Event	Description	Outcome / Status
CBSE National Science Exhibition 2023	National-level exhibition for school science projects conducted by CBSE; project submitted and exhibited in Delhi	Participated — National Level
Dr. APJ Abdul Kalam Ignited Minds Award 2022	Innovation award programme recognising creative student projects	Participated
IdeaNest 2.0 (AIC-NITF)	Startup incubation and idea development bootcamp; covered business modelling, IPR, go-to-market strategy, and investor pitch training	Completed Bootcamp
Sahodaya Science Exhibition	Regional science exhibition conducted by Sahodaya school complex	Participated — Project exhibited
Patent Application (AIC-Nalanda, Batch-13)	Formal patent application filed through AIC Nalanda Innovation Centre as part of IdeaNest 2.0	Application Filed 2022
National IP Awards	National Intellectual Property award application submitted	Application Submitted

13. Conclusions

AGROFILOS represents a practically-motivated engineering solution to one of India's most persistent agricultural challenges: making mechanised pesticide application accessible to the small and marginal farmer. By combining a ruggedised PVC-based telescopic UGV frame with a dual-Arduino RF control system and 7-nozzle sprinkler array, the project demonstrates that effective agricultural automation does not require expensive drones or complex machinery.

The design philosophy throughout has been frugal innovation — maximising utility within tight cost constraints, using widely available materials (PVC pipes, Arduino Nano, off-the-shelf RF modules), and ensuring the system can be operated and maintained by farmers without technical expertise.

Key Achievements

- Designed and built a functional 9-foot telescopic UGV prototype from commercially available materials.
- Achieved reliable RF communication and control across 100+ metres using the RadioHead library.
- Implemented independent pump and motor control with 4 speed levels and 4 flow levels via a compact single-character encoding protocol.
- Integrated battery voltage monitoring, water level sensing, and visual status indicators on both remote and MCU.
- Evolved firmware through 4 major versions; migrated from VirtualWire → Bluetooth → RadioHead RF.
- Presented at national-level competitions including CBSE National Science Exhibition and IdeaNest 2.0 startup bootcamp.
- Filed a formal patent application through AIC-Nalanda Innovation Centre.

Long-Term Vision

AGROFILOS aims to evolve into a commercially viable, mass-produced agricultural UGV platform for the Indian market — priced accessibly enough for small farmers, modular enough to support ploughing, seeding, and fertilization attachments, and smart enough to navigate field boundaries autonomously. The ultimate goal is to remove the farmer from the hazardous chemical environment entirely while dramatically improving spray efficiency and reducing input costs.

14. References & Appendices

14.1 Key References

- Agricultural Census 2015–16, Ministry of Agriculture & Farmers Welfare, Government of India — Small and marginal farmer statistics.
- PMKSY (Pradhan Mantri Krishi Sinchayee Yojana) Guidelines — Government irrigation and mechanisation subsidy framework.
- RadioHead Arduino Library (RH_ASK) Documentation — Mike McCauley, AirSpayce Pty Ltd.
- Arduino Nano v3 Hardware Specification — Arduino.cc.
- Orange DC Motor OG555 Series Datasheet — 12V, 125 RPM, 150 N·cm specifications.
- Cytron MDD3A Dual Channel DC Motor Driver Datasheet — 3A per channel specifications.
- P90NF03L N-Channel Power MOSFET Datasheet — STMicroelectronics.

14.2 Appendix A — Project File Structure

The complete AGROFILOS project is organised in the following directory structure:

Directory / File	Contents
agrofilos.....final.pptx	Final project presentation slides
blueprints/	Engineering drawings: front view, side view, top view, extended view, remote layout
Code/315Mhz RF/RadioHead/	Final firmware: Mains_4.0.ino, Transmitter_4.0.ino (current stable version)
Code/315Mhz RF/VirtualWire/	Previous v2.0 firmware using VirtualWire library
Code/HC-05/	v3.0 Bluetooth firmware (Main_3.0, Transmitter_3.0)
Events/CBSE/	CBSE Science Exhibition documentation and travel bookings
Events/IdeaNest 2.0/	Business model canvas, IPR training materials, GTM strategy documents
Events/Ignite/	Dr. APJ Abdul Kalam Ignited Minds Award submission
Events/Inspire/	INSPIRE award synopsis and guidelines
Events/National IP awards/	National IP Award application forms
Materials/	Component datasheets, schematic references, material images
Patent/	AGROFILOS patent application DOCX, NOC, email correspondence
Script/	Presentation script photographs

14.3 Appendix B — GitHub Repository

Source code for AGROFILOS firmware is available at: github.com/ayusha343

For technical details, collaboration inquiries, or access to full schematics, contact the developer through the GitHub repository or LinkedIn. This documentation represents the state of the prototype as built and self-tested. All specifications are subject to change as the project evolves toward a production design.